

Adroit Technologies

Protocol Driver

Name	Modbus RTU Ethernet TCP/IP Multi Channel
DLL	MOD_EMC



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1. Overview

Adroit supports communication with the following PLCs: Modicon 84, 884, 184, 384, 484, 584, 984, Quantum, Premium, M340, Ethernet/serial gateways and many other third-party devices over TCP/IP using MOD_EMC and its sister driver MOD_ETH (Modbus Ethernet).

MOD_EMC implements various optimizations over MOD_ETH to ensure very high-speed communications and includes the ability to configure multiple channels on one device configuration, scan tasks will then be load balanced over these channels.

MOD_ETH and MOD_EMC are interchangeable due to identical registry and addressing structures, and they can be run simultaneously.

Logical devices are still possible but redundant unless special requirements such as requiring permanent separation of the devices scan list to say stop parts of the list.

2. Revisions

2.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	16-Jan-2023	K. Robinson	Grammar edits and content additions.	D. Jordaan
2.0	15-Jan-2024	D. Jordaan	Doc updated sync with Rev 132	-
3.0				
4.0				
5.0				
6.0				

2.2. DLL Revision

Rev.	Date	Changes
111	23-Oct-2014	First release.
130	24-Nov-2023	UK version compatible.
131	21-Dec-2023	CRITICAL FIX ... 30000x L,F,C and S types would fetch from 4x Holding register areas
130	15-Jan-2023	Markbad cleaned up, Dynamic config on start/stop, Added Backoff mechanism to match MOD_ETH

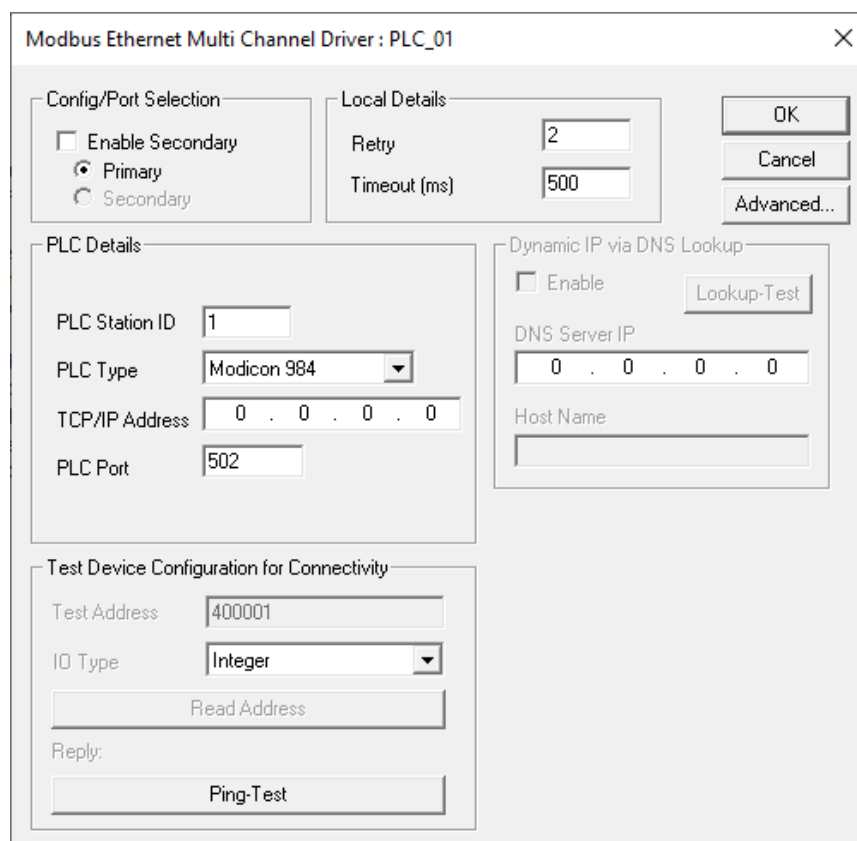
3. Wiring

Standard Ethernet interconnection.

4. Device Configuration

4.1. Device Configuration

Please note: the registry is only updated upon pressing the final ok button on the main device config dialog.



The image shows a configuration dialog box titled "Modbus Ethernet Multi Channel Driver : PLC_01". It contains several sections:

- Config/Port Selection:** Includes radio buttons for "Enable Secondary" (unchecked), "Primary" (selected), and "Secondary".
- Local Details:** Includes input fields for "Retry" (set to 2) and "Timeout (ms)" (set to 500).
- PLC Details:** Includes input fields for "PLC Station ID" (set to 1), "PLC Type" (set to Modicon 984), "TCP/IP Address" (set to 0.0.0.0), and "PLC Port" (set to 502).
- Dynamic IP via DNS Lookup:** Includes a checkbox for "Enable" (unchecked) and a "Lookup-Test" button. Below it are input fields for "DNS Server IP" (set to 0.0.0.0) and "Host Name".
- Test Device Configuration for Connectivity:** Includes input fields for "Test Address" (set to 400001) and "IO Type" (set to Integer). It also has buttons for "Read Address", "Reply:", and "Ping-Test".

Buttons for "OK", "Cancel", and "Advanced..." are located in the top right corner.

4.1.1. Config/Port Selection

Enable Secondary – (Default = unchecked) Enables secondary configuration for dual redundant PLCs, via **Primary** and **Secondary** radio buttons. (Do not enable if a secondary channel is not configured correctly.) Select the option using the radio buttons to configure each configuration.

4.1.2. PLC Details

PLC Station ID - Valid Station number. See section [Protocol Definition and Considerations - 6.1 Address Padding](#) for definition of address padding for addresses greater than 254.

PLC Type - Applicable PLC types.

PLC IP/Address – The TCP/IP address of the PLC.

PLC Port Address – (Default = port 502) The port address of the PLC.



PLEASE NOTE

For Multiple logical mappings to one PLC (as multiple Adroit devices), use the same IP and Port number. The driver will only open multiple sockets (parallel connections) to this physical device if the "Create multiple connections..." checkbox is selected in the [Advanced Settings Dialog](#).

4.1.3. Dynamic IP via DNS Lookup (Disabled until required)

When Dynamic IP via DNS is enabled, the driver will launch a thread that will attempt to lookup an IP address when retries are exhausted and found to be failing. This lookup process will continue indefinitely until the Agent Server is stopped. It will not block a connection or the Agent Server read/writes which will continue on existing connection parameters until changed by a successful lookup, which will then save those new parameters.

This mode can be used on connections that are on the internet and have a dynamic IP such as when using ADSL that change IP address once a day. The device will need to be registered on a DNS server such as DYNDNS.ORG

Enable – (Default = unchecked) This will enable the DNS Lookup mechanism.

DNS Server IP – IP address of the DNS server that will handle lookup requests.

Host Name – The name of the machine/PLC registered on the DNS server. This name is used to request the associated IP address. The IP address returned will be stored in the IP Address field dynamically in the PLC Details.

4.1.4. Local Details

Retry - The number of attempts that the Master will attempt to communicate with the PLC before failing a transaction.

Timeout - The time that the Master will wait for a valid response from a PLC before failing an individual transaction (Read or Write).

4.1.5. Test Device Config for Connectivity

This section is available to help users determine whether the settings they have entered are valid and Adroit can communicate with your device. The Agent Server is not required to be running.

Test Address – An address in your device which will be polled when the "Read Address" button is clicked. The address entered must be valid and will match the IEC or Standard Modbus addressing scheme selected in the advanced dialog.

IO Type – Defines the base address type to be read, i.e., Boolean, Integer, Real or String80.

Read Address – Click this button to poll the PLC with the current settings in the dialog.

Reply – The successful or failed result will be updated to this static text control.

Ping Test – Click to verify that the TCP/IP address entered into the boxes can be accessed using a ping, an appropriate message box will appear as to the result.

4.2. Advanced Settings Dialog

Advanced Button – Access advanced communication parameters and driver settings for this device.



PLEASE NOTE

Please note that the registry is only updated upon pressing the final OK button on the main device configuration dialog



INFORMATION

For all configuration settings: these are offline changes to the registry and are only reloaded at start-up (hence we suggest a restart of the Agent Server). There is a case where after a read completely timeouts and exceeding all the retries as well, the driver will dynamically reload the registry and a restart is then not required. This may not be entirely apparent so when in doubt it is suggested to restart the Agent Server regardless just to make sure.

Advanced Modbus Ethernet settings
✕

Please read the driver documentation before modifying any of these settings.
 Default values '=' are shown in brackets.
 For Backoff and Tolerance modes a consecutive fail count increments only after the poll attempt AND its retries fail.

OK

Cancel

Help

Protocol delays

TX-RX delay between any TX and RX required on some devices where responses arrive in multiple packets. (mSec) (0=off) (default = 0) (Capped at 500mSec Max) 0

Backoff period (mSec) (0=off) (default = 20000) 10000

Backoff count (only backoff after this number of consecutive failures) (default = 2) 2

Byte Ordering

☐ Reverse Modbus coil MSB/LSB ordering when scanning them into integers (default = off)

☒ Force 16Bit boundary justification. (Default = on)

☐ Reverse MSb/LSb ordering when scanning Holding register bits into Booleans. (default = off)

☐ Swap byte order on String agents. (default = on)

☐ Swap word's byte order. (default = on)

☐ Swap LONG word ordering (default = on)

☐ Swap floating-point word ordering (default = off)

General

☒ Use sequence numbering in the M-BUS header (on=greater integrity during timeouts) (default = on)

☒ Optimize scanning of coils and relays to only scan the exact number of elements (default = off)

☐ Use Function6 (Write-Single Holding Register) (default = off)

☒ Insert spaces for non printable chars in strings (default = on)

☐ Use IEC Standard Addressing (default = off)

☒ Address 255 Padding Required (default = on)

☒ Detailed Debbugging information (default = off)

☐ Use Modbus serial encapsulated in TCP/IP (default = off)

Connection

☒ Close port if all devices on a comres is stopped (default = on)

☐ Close Port after every transaction to the slave (default = off)

☐ Close and re-connect to the slave on all communication errors (off=only close when retries exhausted) (default = off)

The number of channels to use when connecting the a device. (use with caution make sure the plc allows this number of simultaneous connections.) (Default = 1, Max = 8) 1

Custom Plc Type :: PDU lengths

Coils 00000	(Bits) (Recommend groups of 8)	1984
Inputs 10000	(Bits) (Recommend groups of 8)	1984
Analogue 30000	(Words) (Recommend at least 2)	124
Holding 40000	(Words) (Recommend at least 2)	2
Extended 60000	(Words) (Recommend at least2)	123

Note: To use these custom pdu sizes - CUSTOM model must be selected in the config dialogue. Agent server MUST be restarted for job sizes to change.

4.2.1. Protocol Delays

TX-RX Delay – (0 to disable) will apply a latency delay between a request and receive attempt. (used if there is a permanent delay required to give an device to process.)

Backoff period (mSec) (0=off) (Default=20000) – (0 to disable) After the backoff-count is consecutively exceeded the driver will enter a backoff mode, rejecting any further reads or writes on all channels until the period is exceeded, then one is allowed through to test for resumption , if it still fails then the driver will re-enter the backoff period again.

Backoff count (mSec) ... (Default=2) After the backoff-count is consecutively exceeded the driver will enter a backoff mode, rejecting any further reads or writes on all channels until the period is exceeded, then one is allowed through to test for resumption , if it still fails then the driver will re-enter the backoff period again.



PLEASE NOTE

Please note setting backoff count to too high a value will negate the usefulness , recommended < 10 , remember timeouts and retries are also taken into account

4.2.2. Byte Ordering

Reverse Modbus coil MSB/LSB ordering when scanning them into integers – If Integer or Real Agents are scanned to coil(0xxxxx) or input(1xxxxx) region and the result appears jumbled, use this setting to reverse the ordering and swap the endian order.

Force 16bit boundary justification. - If (checked - default) the driver will force Integer or Real Agents scanned to coil(0xxxxx) or input(1xxxxx) region to be mapped into 16 bit word boundaries.

Reverse MSByte/LSByte ordering when scanning Holding register bits into Booleans. - before unpacking to Boolean Agents the word data returned from a holding register is first byte swapped.

Swap byte ordering on Strings Agents – Toggle to change the byte order of each word that the string occupies, for example "EHLL10" --> "HELLO1" .

Swap word's byte order– Toggle to change the byte order of word 16 bit register results.

Swap LONG word ordering – Toggle to change the word order of long 32 bit register results.

Swap floating-point value ordering – Toggle to change the word order of floating 32 bit register results. Must be selected for communication to all genuine Modicon equipment.

4.2.3. General

Use sequence numbering in the M-BUS header – (Default = checked) increases data integrity if using a slow and noisy media or getting timeouts. This option may be incompatible with simple Modbus slaves which do not copy the frame reference bytes in the open-MBUS header.

Optimize scanning of Coils and relays to only scan the exact number of elements – turn on this option to prevent the driver from reading multiples of 8 bits (at least 1 byte), and instead read only the required number of bits.

Use function6 (Write Single Holding Register) – (Default = unchecked) This will use function 06 to write single registers, instead of writing one register using multiple register write function, some PLCs only support function 06.

Insert spaces for non-printable characters in strings – (Default = checked) This will change any character that is less than a space " " 0x20 and greater than "~" 0x7e. These are referred to as

non-printable and can cause load and save problems in the wgp file. (Opt-out if previous functionality is needed).

Detailed debugging information – (Default = unchecked) This causes the driver to add more debugging information to the datascope, such as data values and more specific information relating to each protocol message.

Use IEC Standard Addressing - (Default = checked) If checked it will enable the driver to use IEC addressing when validating, for example %MW0 instead of 400001. %MW0 = 400001.

Address 255 Padding required - (Default = checked) If checked it will force the driver to use a 16-bit address word prefixed with a 0xFF pad byte when the address is larger than 254 decimal value.

See **Address padding** below for examples.

Use Modbus serial encapsulation in TCP/IP - (Default = unchecked) If checked it will remove the TCP header and add a CRC to match the serial Modbus protocol. This data is encapsulated into the data portion of the IP packet and sent to a dumb TCP to serial convertor/gateway where the data portion is simple retransmitted on its serial bus.



PLEASE NOTE

Some gateways are intelligent and do not need the driver to prepare the Modbus packet. You may need to investigate the device specifications to see if it prepares this for you, or you may need to experiment with this setting enabled and disabled.

4.2.4. Connection

Close Port if all devices on a comres is stopped (default = on) – The socket will be closed if all devices sharing the communications resource object are stopped.

Number of channels... – (Default = 1, Max = 8) Used to configure multiple channels on one device configuration. Scan tasks will then be load balanced over these channels via a single scanning list. Logical devices are still possible but redundant unless there are special requirements such as stopping scanning on certain plant information whilst having communications on another area.

Close Port after every transaction to the slave – This setting should only be used in order to keep the connection closed after successful communication, typically used on GPRS networks that will close the connection anyway, now the driver will expect it and re-open the connection when it wants to communicate again.

Close and re-connect to the slave on all communication errors – This setting should only be used to force the driver to reset the connection to the host in between every failed attempt and or retry. If this setting is unchecked the driver will behave normally, only resetting the connection once after complete failure and all associated retries are exhausted.

4.2.5. Custom PLC Type: PDU Lengths



PLEASE NOTE

This is an advanced feature. Changing these settings without special consideration can break the communication altogether.

Used in conjunction with the "CUSTOM" PLC type, set the appropriate limits here. If the device type is anything other than custom, the custom values will be ignored. Length count is in bits for Outputs (coils) and inputs, and 16-bit registers for the others.

Outputs 000000, Inputs 100000 -It is best to have length in multiples of 8, i.e., 16 will result in 2 data bytes returned in the data portion of the PDU. Some bit-devices do not respond correctly if not using a multiple (a minimum of 1 and maximum of 1984).

Analogue 300000, Holding 400000, Extended holding 600000 - 16-bit register types; any number can be used (a minimum of 2 and maximum of 124 registers).



PLEASE NOTE

Analog and Long data types require two Modbus 16-bit registers per item. So, if these are used, make sure the PDU's can contain an even number of items.

5. Supported Addresses

5.1. Using Modbus Addressing:

Device		Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Outputs	1	000001	065535	Digital	000001			
	1			Decimal		000001	000001	
	2			Signed Integer		000001 I	000001 I	
	3			BCD16		000001 B	000001 B	
Inputs	4	100001	165535	Digital	100001			
	4			Decimal		100001		
	5			Signed Integer		100001 I	100001 I	
	6			BCD16		100001 B	100001 B	
Input Registers	7	300001	365535	Digital	300001.1			
	7			Decimal		300001	300001	
	8			Signed Integer		300001 I	300001 I	
	9			BCD16		300001 B	300001 B	
Holding Registers	10	400001	465535	Digital	400001.1			
	10			Decimal		400001	400001	
	11			Signed Integer		400001 I	400001 I	
	12			BCD16		400001 B	400001 B	
	13			Long Decimal		400001 L	400001 L	
	14			Float			400001 F	
	15			Pascal-String				400001 S20
	16			Char String				400001 C20
Extended Registers	17	600000	698303	Digital	600000.1			
	17			Decimal		600000	600000	
	18			Signed Integer		600000 I	600000 I	
	19			BCD16		600000 B	600000 B	
	20			Long Decimal		600000 L	600000 L	
	21			Float			600000 F	
	22			Pascal-String				600001 S20
	23			Char String				600000 C20

5.2. Using IEC Addressing :

Device		Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Outputs	1	0	65535	Digital	%Q0			
	1			Decimal		%Q0	%Q0	
	2			Signed Integer		%Q0 I	%Q0 I	
	3			BCD16		%Q0 B	%Q0 B	
Inputs	4	0	65535	Digital	%I0			
	4			Decimal		%I0		
	5			Signed Integer		%I0 I	%I0 I	
	6			BCD16		%I0 B	%I0 B	
Input Registers	7	0	65535	Digital	%IW0.1			
	7			Decimal		%IW0	%IW0	
	8			Signed Integer		%IW0 I	%IW0 I	
	9			BCD16		%IW0 B	%IW0 B	
Holding Registers	10	0	65535	Digital	%MW0.1			
	10			Decimal		%MW0	%MW0	
	11			Signed Integer		%MW0 I	%MW0 I	
	12			BCD16		%MW0 B	%MW0 B	
	13			Long Decimal		%MW0 L	%MW0 L	
	14			Float			%MW0 F	
	15			Pascal-String				%MW0 S20
	16			Char String				%MW0 C20
Extended Registers	17	0	98303	Digital	%ME0.1			
	17			Decimal		%ME0	%ME0	
	18			Signed Integer		%ME0 I	%ME0 I	
	19			BCD16		%ME0 B	%ME0 B	
	20			Long Decimal		%ME0 L	%ME0 L	
	21			Float			%ME0 F	
	22			Pascal-String				%ME0 S20
	23			Char String				%ME0 C20

Note 1: Writes are not supported to inputs 100001 through to 165535, and Input registers 300001 through 365535 as these are read only mappings.

Note 2: Unsolicited support for Holding registers (400001) region only. When the PLC sends unsolicited data only those holding registers that are mapped into Adroit will be updated if they are not mapped Adroit will ignore those values.

Note 3: Char strings do not use a prefix length register so each register holds two characters and can be considered to be a simple array of word registers each containing two characters. Pascal strings are prefixed by 1 register to hold the actual length of the string. This is NOT the same as the scanning address length which indicates the maximum length.



WARNING

If 000001 (Output) region is mapped as an Integer or Real for say a Marshal Agent, you will be using function "0x15" – force multiple coils, which will write 16 coils at once, possibly overwriting some unintentionally. That means you can inadvertently control the plant. Make sure you are aware of this before mapping coils to Integers.

See the [Advanced Settings Dialog](#) for details on swapping byte order for this type of mapping to coils and inputs.

5.3. Third Party Modbus Emulators

A number of hardware manufacturers have adopted the Modbus protocol to communicate to their hardware. These devices use various hardware interfaces that have different maximum buffers and thus number of points that can be scanned in one scan task varies on available buffer sizes not withstanding protocol limitations. Since Adroit creates scan tasks/jobs automatically, it is important to choose a PLC type that would not result in scan task requesting for more data in one scan than that can be handled by the protocol.

Consult the list below to choose the most appropriate PLC Type.

From revision 25 onwards there is a "CUSTOM" CPU type selection where the PDU lengths are individually configurable in the [Advanced Settings Dialog](#).

The maximum number of items (Digital bits or 16-bit registers) that can be communicated for the different Modicon PLC's is shown below:

	84	884	184	384	484	584	984
000001/%Q0	64	1984	800	800	512	1984	1984
100001/%I0	64	1984	800	800	512	1984	1984
300001/%IW0	4	124	100	100	32	124	124
400001/%MW0	32	124	100	100	124	124	124
600000/%ME0	32	123	100	100	123	123	123

	Custom	Quantum	Premium	M340
000001/%Q0	*	1984	1984	1984
100001/%I0	*	1984	1984	1984
300001/%IW0	*	124	124	124
400001/%MW0	*	124	124	124
600000/%ME0	*	123	123	123

If the PLC slave is not one of those in the list, and is emulating Modbus, use this table to determine a match for the maximum telegram lengths supported by PLC. Selecting shorter telegram lengths produces slower performance due to more scan tasks, but ensures compatibility.

* PDU lengths based on vendor-supplied information see [Advanced Settings Dialog](#).

6. Protocol Definition and Considerations

6.1. Address Padding

Normal addresses from 1 – 254, ID occupies 1 byte.

Slave/Unit ID
01

For addresses from 255 – 65535 the address is 'padded' with 0xFF and the ID now occupies two bytes.

Padding	Slave Hi ID	Slave Lo ID
FF	xx	xx

6.1.1. TCP/IP Header Structure (Prefix structure for ALL messages)

Note: There is no CRC field present in the TCP/IP data frame, message integrity is catered for in the TCP/IP protocol and connection.

TCP header:

Transaction Hi ref	Transaction Lo ref	Protocol ID	Protocol ID	Hi Len	Lo Len	Slave/Unit ID
00	00	00	00	00	xx	01

Transaction reference number field is used since revision 34. Please see the [Advanced Settings Dialog](#) for more detail. The unit ID (in grey) is shown here as part of the header for illustration purposes only. Some devices may ignore the unit ID or station-# field or use a default of 1.

6.1.2. Coils 000001- 065535 (%Q0-%Q65535)

Read multiple coils 01 request:

Function	Start Addr Hi	Start Addr Lo	Bit count Hi	Bit count Lo
01	00	01	00	01

Read response:

Function	Byte Count	Data
01	01	01

Exception error response: (see table below for error definition)

Function	Code
81	02

Example: Read 1 coil at reference 0 (000001) resulting in value 1.

Write single coil 05

Write request single:

Function	Coil Addr Hi	Coil Addr Lo	Force Data Hi	Force Data Lo
05	00	00	FF	00

Write response:

Function	Coil Addr Hi	Coil Addr Lo	Force Data Hi	Force Data Lo
05	00	00	FF	00

Exception error response: (see table below for error definition)

Function	Code
85	02

Example: Write 1 coil at reference 0 (000001) resulting in 0 -> 1
(FF00 = true/on)(0000 = False/off)

Write Multiple / Force multiple 0F (i.e. Coils mapped as integer):

Function	Coil Addr Hi	Coil Addr Lo	Bit count Data Hi	Bit count Data Lo	Byte Count	Force Data Hi	Force Data Lo
0f	00	00	00	0f	01	12	34

Write response:

Function	Coil Addr Hi	Coil Addr Lo	Bit count Data Hi	Bit count Data Lo
0f	00	00	00	0f

Exception error response: (see table below for error definition)

Function	Code
8f	02

Example: Write 16 coils at reference 0 (000001) through (000016) resulting in
<- MSB 00000000001111100 -> LSB -- note warning above.

6.1.3. Inputs 100001- 165535 (%I0-%I65535)

Read relays 02 request:

Function	Start Addr Hi	Start Addr Lo	Bit count Hi	Bit count Lo
02	00	00	00	01

Read response:

Function	Byte Count	Data
02	01	01

Exception error response: (see table below for error definition)

Function	Code
82	02

Example: Read 1 discrete input at reference 0 (100001) resulting in value 1

Write relays request:

Not supported in Modbus protocol, and normally it is not possible to write to an input.

6.1.4. Input Registers 300001- 365535 (%IW0-%IW65535)

Read request:

Function	Start Addr Hi	Start Addr Lo	Word count Hi	Word count Lo
04	00	00	00	01

Read response:

Function	Byte Count	Data Hi	Data Lo
04	02	12	34

Exception error response: (see table below for error definition)

Function	Code
84	02

Example: Read 1 input register at reference 0 (300001) resulting in value 1234 hex

Write request:

Not supported in Modbus protocol

6.1.5. Holding Registers 400001- 465535 (%MW0-%MW65535)

Read multiple holding registers 03 request:

Function	Start Addr Hi	Start Addr Lo	Word count Hi	Word count Lo
03	00	00	00	01

Read response:

Function	Byte Count	Data Hi	Data Lo
03	02	12	34

Exception error response: (see table below for error definition)

Function	Code
83	02

Example: Read 1 register at reference 0 (400001) resulting in value 1234 hex

Write holding register 10 request:

Function	Start Addr Hi	Start Addr Lo	Word count Hi	Word count Lo	Byte count	Data Hi	Data Lo
10	00	00	00	01	02	12	34

Write response:

Function	Start Addr Hi	Start Addr Lo	Word count Hi	Word count Lo
10	00	13	00	02

Exception error response: (see table below for error definition)

Function	Code
90	02

Example: Write 1 register at reference 0 (400001) to 1234 hex

Write/Preset single register 06 request:

Not shown, this frame (command 06) will be used in preference to the Write multiple registers (command 09) only when writing a single 16-bit tag, and if the option to do so is turned on.

6.1.6. Extended Registers 600001- 498303(NB 600000 is valid) (%ME0-%ME98303)

Read request:

Function	Remainder Count	Register Area	File Addr Hi	File Addr Lo	Offset Addr Hi	Offset Addr Lo	Word count Hi	Word count Lo
14	07	06	00	01	00	02	00	01

Read response:

Function	Overall Count	Byte Count	Register Area	Data Hi	Data Lo
14	04	03	06	12	34

Exception error response: (see table below for error definition)

Function	Code
94	02

Example: Read 1 register at reference 1, 2 (File 1 offset 2: 600001) resulting in value 1234 hex (Note: File 1, Offset 0 = 600000)

Write multiple registers 09 request:

Function	Remainder Count	Register Area	File Addr Hi	File Addr Lo	Offset Addr Hi	Offset Addr Lo	Word count Hi	Word Count Lo	Data hi	Data Lo
15	09	06	00	01	00	02	00	01	12	34

Write response:

Function	Remainder Count	Register Area	File Addr Hi	File Addr Lo	Offset Addr Hi	Offset Addr Lo	Word count Hi	Word Count Lo	Data hi	Data Lo
15	09	06	00	01	00	02	00	01	12	34

Exception error response: (see table below for error definition)

Function	Code
95	02

Example: Write 1 register at reference 1,2 (File 1 Offset 2 : 600001) to 1234 hex

Note: File 1, Offset 0 = 600000

7. General Notes and Troubleshooting

7.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

7.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7.3. PLC Exception Codes

These codes are returned by the PLC. Some action either in scan configuration, driver setup parameters or hardware configuration may be needed to rectify the problem.

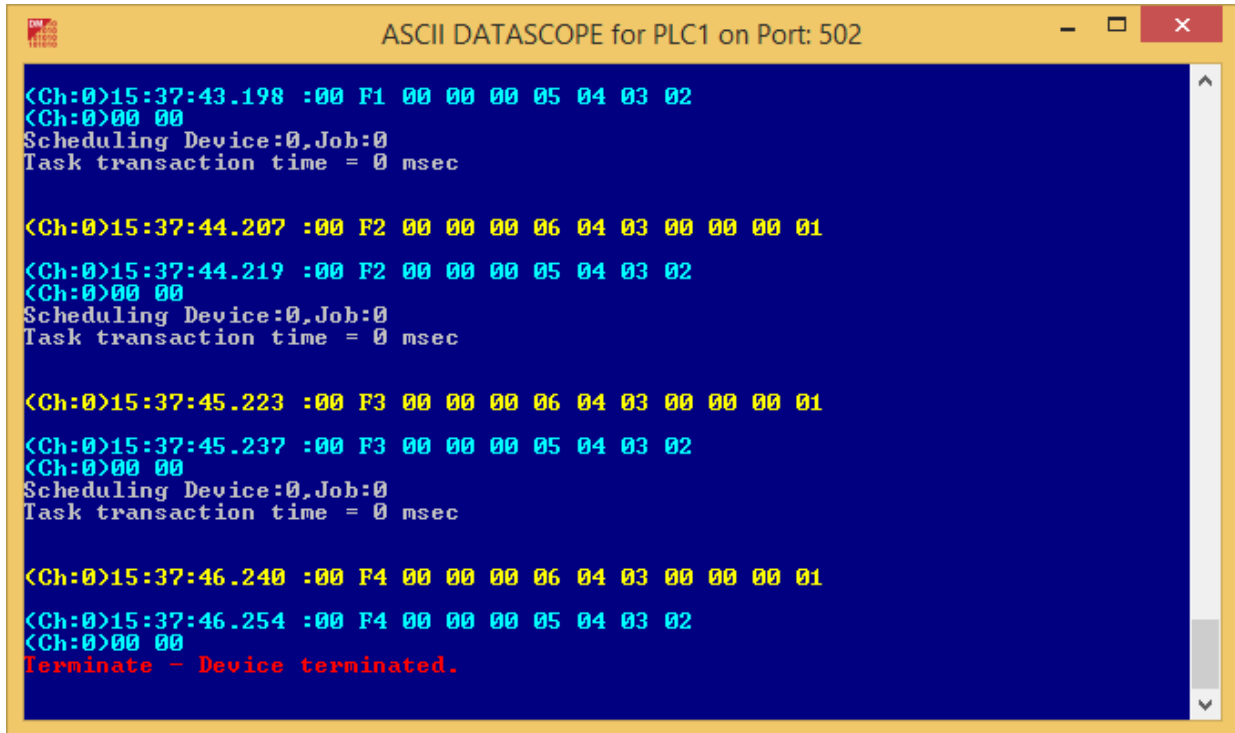
Hex code	Meaning	Description
0x0	No error	
0x1	Illegal Function	The function code received in the query is not an allowable action for the slave.
0x2	Illegal Data Address	The data address received is not an allowable address for the slave.
0x3	Illegal Data value	A value contained in the query data field is not an allowable value for the slave.
0x4	Slave Device failure	Unrecoverable error occurred in the slave while processing the request.
0x5	Acknowledge	This response is returned if the slave is processing the request but will take a long time; this is to prevent the master from timing out.
0x6	Slave device busy	Re-transmit the command later when the slave is not busy.
0x7	Negative acknowledge	Slave cannot perform a programming function requested.
0x8	Memory Parity error	The slave may require service. This error may occur when a request is made to read extended memory and a parity error is detected in the memory.
0xA	Gateway Path un-available	The remote device on a Modbus plus + network did not respond to the gateway in time, this may occurs if a device routes through the gateway onto a different network and is no longer connected.
0xB	Gateway Target failed to respond.	The remote device did not respond to the gateway in time, this occurs if a device routes through a gateway onto a different network such as RS485 and is no longer connected.

7.4. Diagnostic Traces

Action	State	Meaning
SEQ word error, nnnn, expected nnnn	Response was out of sequence	The reference filed in the header of the TCP frame was not correctly copied by the slave. This occurs if a slave responds too late to our request, and hence goes out of sync. If you see this message and get no comms at all, you may want to turn off the frame sequence numbering option in the advanced settings.
Cannot create file handle for driver monitoring	Driver not operational	1. Check that you have at least 500kb free disk space. 2. Check that you have file create rights on the working-directory. 3. A folder with the same name as the Device Agent exists in the working directory.
Invalid BCD value from PLC (max value = 9999)	Driver not able to parse return message from PLC	A request for a range of BCD values included some non-valid entries; try ensuring all BCD data's are packed sequentially into registers, and that they are all valid.
NNN bytes of noise purged	Noise Purged OK	Data was received unexpectedly over the connection; this may occur after a timeout if the slave responds too late.
PLC responded with exception - zzzzzz	Driver will retry	See the table above for a description of the exception codes. These codes are codes returned by the PLC, not driver errors.
Successful write to device/ Successful read from device	Driver healthy.	The driver has removed from a error, either of these 2 messages will appear only once following any prior error.
Primary Socket initialised; Secondary Socket not initialized	Device zzzz healthy	This 'informational' event will appear in the logs at start-up and indicates that the Windows TCP/IP protocol stack is in working order.

7.5. Datascope Considerations

A channel number will be prepended to the datascope. Channels can send data to the scope in any order , interested parties need to keep track of the channel number.



```

ASCII DATASCOPE for PLC1 on Port: 502

<Ch:0>15:37:43.198 :00 F1 00 00 00 05 04 03 02
<Ch:0>00 00
Scheduling Device:0,Job:0
Task transaction time = 0 msec

<Ch:0>15:37:44.207 :00 F2 00 00 00 06 04 03 00 00 00 01
<Ch:0>15:37:44.219 :00 F2 00 00 00 05 04 03 02
<Ch:0>00 00
Scheduling Device:0,Job:0
Task transaction time = 0 msec

<Ch:0>15:37:45.223 :00 F3 00 00 00 06 04 03 00 00 00 01
<Ch:0>15:37:45.237 :00 F3 00 00 00 05 04 03 02
<Ch:0>00 00
Scheduling Device:0,Job:0
Task transaction time = 0 msec

<Ch:0>15:37:46.240 :00 F4 00 00 00 06 04 03 00 00 00 01
<Ch:0>15:37:46.254 :00 F4 00 00 00 05 04 03 02
<Ch:0>00 00
Terminate - Device terminated.
  
```